

GitHub Actions Workflow

COMS W4156: Advanced Software Engineering

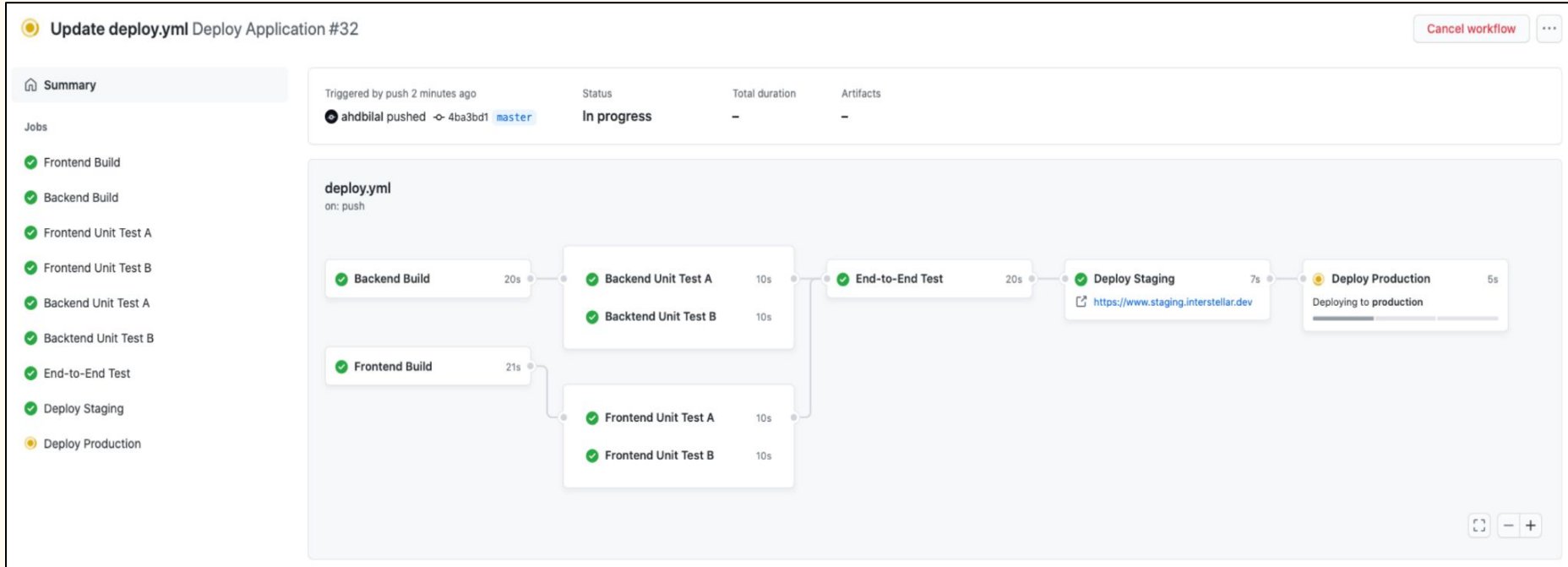
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What is it?

Why do we use it?

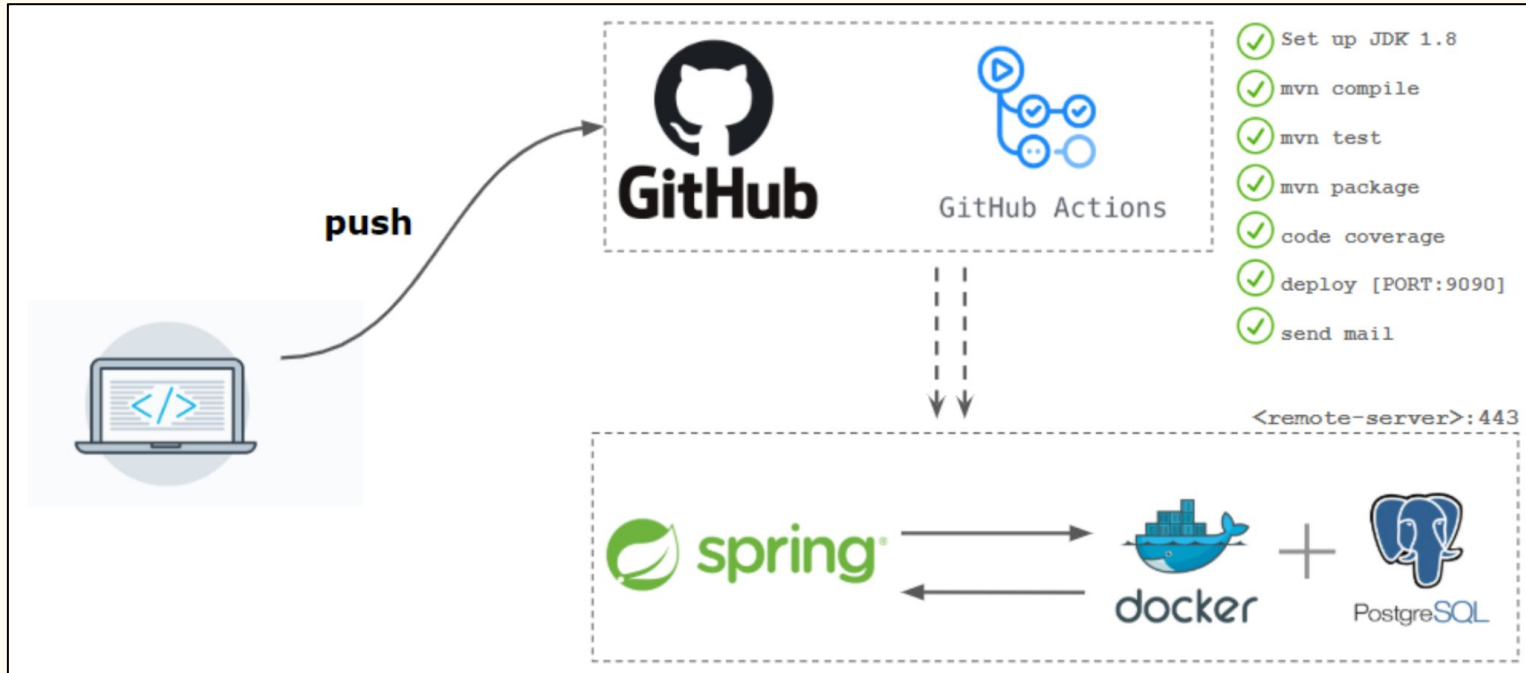
- Tool to automate repetitive and time-consuming tasks in your software development workflow
- **Continuous Integration:** Can automatically build and test your code when you push to GitHub, ensuring that it remains functional and bug-free
- **Continuous Deployment:** Allows you to seamlessly deploy your application when code changes are merged
- Used for a wide range of tasks beyond just CI/CD.
Eg: automating code formatting, generating documentation, notifying your team of events

Workflows Example - Simple



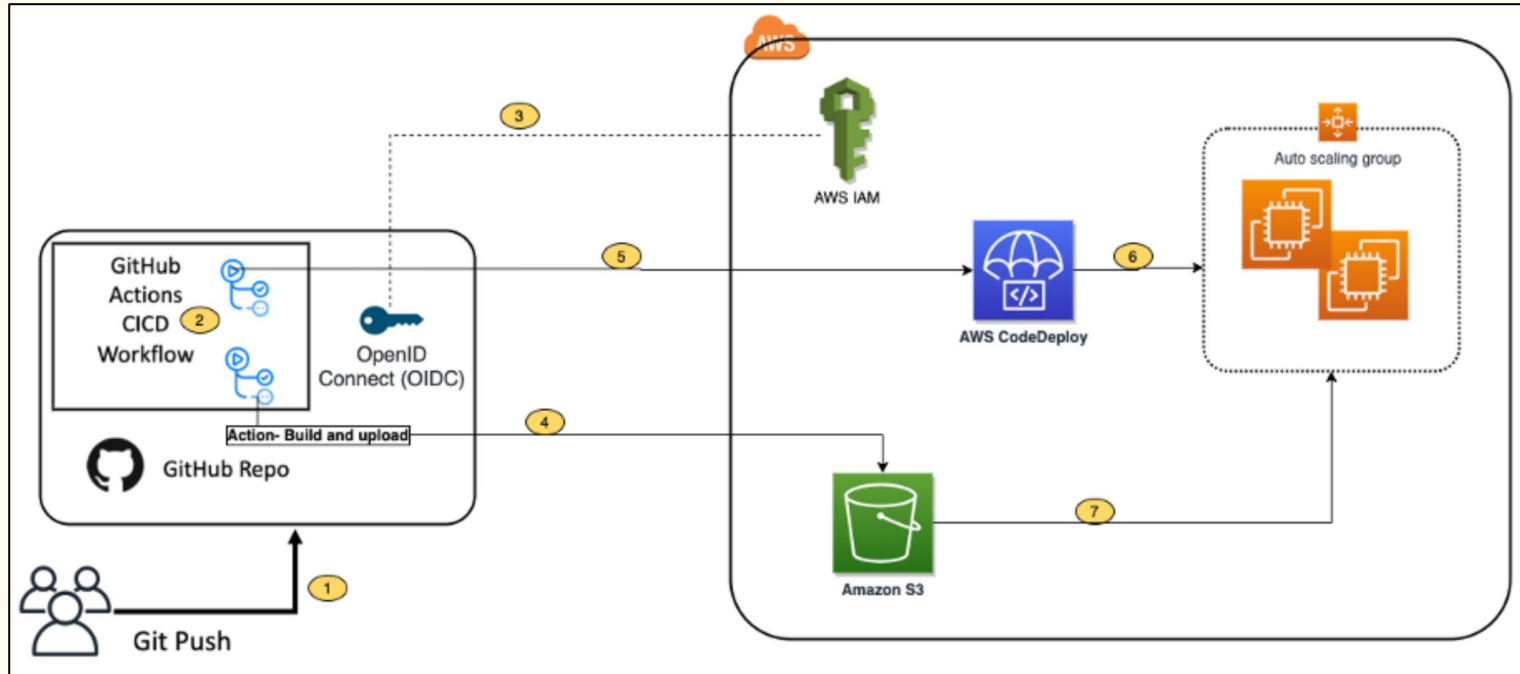
<https://github.blog/changelog/2020-12-08-github-actions-workflow-visualization/>

Workflows Example - More Complex



<https://dzone.com/articles/automate-your-development-workflow-with-github-act-1>

Workflows Example - Very Complex

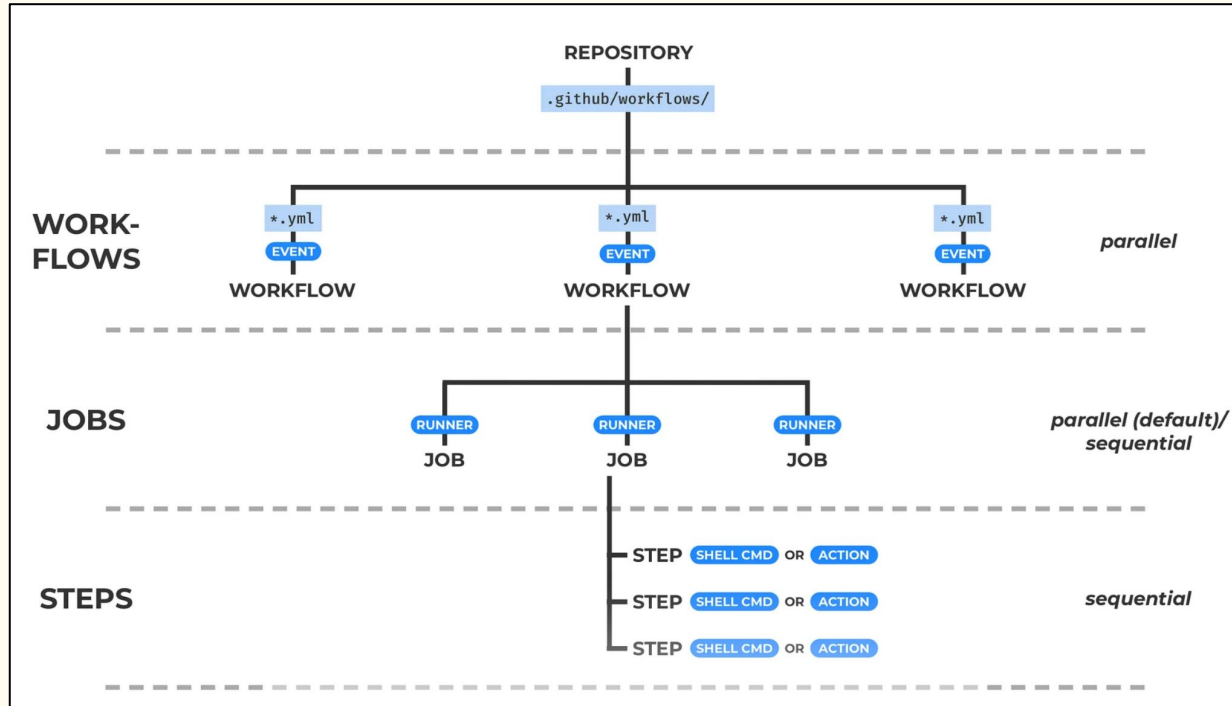


<https://aws.amazon.com/blogs/devops/integrating-with-github-actions-ci-cd-pipeline-to-deploy-a-web-app-to-amazon-ec2/>

Core Concepts

- **Event:** A specific activity in a repo that may trigger a workflow.
 - A workflow can be triggered by one or more events.
 - Eg: New commit/pull request
 - Types of events: Internal, external (repository_dispatch event), scheduled, manual
- **Action:** Application which performs a complex but frequently repeated task.
 - Custom or from GitHub Marketplace
 - Eg: Pull your git repository from GitHub, set up the correct toolchain for build environment
- **Step:** An executable component, could be a shell script or an Action
- **Job:** A series of steps in a workflow that execute on the same server (“runner”).
 - Can configure dependencies between jobs
- **Workflow:** A configurable automated process that runs multiple jobs.
 - Defined by a YAML/YML file in `.github/workflows`
 - A repo can have multiple workflows

Core Concepts - Visualization



<https://itnext.io/getting-started-with-github-actions-fe94167dbc6d>

Create Workflow - YAML File

```
name: ASE Bank App CI using Java Maven
```

-----> Name of the workflow [optional]

```
on:
```

```
  push:
```

```
    branches: [ "main" ]
```

```
  pull_request:
```

```
    branches: [ "main" ]
```

-----> Events to trigger the workflow

```
jobs:
```

-----> List of all jobs in the workflow

```
  build:
```

```
    name: Set up and run tests
```

```
    runs-on: ubuntu-latest
```

-----> Job will run on the latest version of Ubuntu runner

```
  steps:
```

-----> List of all the steps in the job “build”

```
- uses: actions/checkout@v3
```

-----> Checkout v3 action: checks out repo onto runner

```
- name: Set up JDK 17
```

```
  uses: actions/setup-java@v3
```

-----> Installs the specified version of the Java

```
  with:
```

```
    java-version: '21'
```

```
    distribution: 'temurin'
```

```
    cache: maven
```

```
- name: Run tests with Maven
```

```
  run: mvn test
```

-----> Executes this command in the runner

Demo

- Will work with a simple Banking interface project
- Technologies: Maven, Java
- <https://github.com/RishabhS66/Github-Actions-Demo>

Comparisons with other CI/CD Tools

Pros:

- Tightly integrated with GitHub, straightforward to use instead of third-party integrations
- Setting up the pipeline is easy and intuitive, don't need to maintain your own servers or infrastructure
- Lesser DevOps expertise and time consumed: built for Developers

Cons:

- Super young, launched in 2018; not as robust with community support as, say, Jenkins
- For Private codebases, need to pay for runner minutes to ensure workflows execute

Resources

- <https://docs.github.com/en/actions/using-workflows/about-workflows>
- <https://docs.github.com/en/actions/learn-github-actions/understanding-github-actions#the-components-of-github-actions>
- <https://docs.github.com/en/actions/using-workflows/events-that-trigger-workflows>
- <https://github.com/actions/starter-workflows>



Workflow Run Complete!